

TEST BANK

CHAPTER 9

TROPICAL COMMUNITIES

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A and then to place the letter for the selected choice from column B in the space provided to the left of the corresponding item in column A.

[Note to instructors: Here, the choices are correctly matched and thus should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. ahermatypic	—————>	A. non-reef-forming corals
___ 2. hermatypic	—————>	B. reef-forming corals
___ 3. developed by Darwin	—————>	C. subsidence theory
___ 4. nematocysts	—————>	D. stinging organelles
___ 5. atolls	—————>	E. ring-shaped reefs
___ 6. barrier reefs	—————>	F. long reefs adjacent to land
___ 7. sweeper tentacles	—————>	G. tools of interclonal aggression
___ 8. soft coral	—————>	H. sea whips and fans
___ 9. crinotoxin	—————>	I. fish toxin extruded from the body
___ 10. ciguatera	—————>	J. disease linked to eating certain fish
___ 11. mangal	—————>	K. mangrove forest
___ 12. pneumatophores	—————>	L. shallow root extensions
___ 13. vivipary	—————>	M. uninterrupted embryonic growth
___ 14. pioneering species	—————>	N. red mangrove
___ 15. prop roots	—————>	O. extensions from branches

Multiple Choice

Circle the choice that best completes or answers the given questions.

[N.B.: The correct answer to each of the questions is indicated by an asterisk. For most of the questions, the letter “a” is the correct choice; reorder the choices as you see fit.]

- | | |
|--|--|
| <p>1. To which taxonomic group do the non-reef-forming corals belong?</p> <ul style="list-style-type: none"> a. Cnidaria * b. Anthozoa c. Gastropoda d. Porifera | <p>2. To which of the following organisms are corals most closely related?</p> <ul style="list-style-type: none"> a. Sea anemones * b. <i>Hydra</i> c. Jellyfish d. Sea cucumbers e. Snails |
|--|--|

166 Test Bank for Marine Biology

3. Coral polyps occupy small cups called _____ in a massive calcium carbonate skeleton.
 - a. corallites *
 - b. sclerosepta
 - c. nematocysts
 - d. gastrovascular cavities
 - e. zooxanthellae
4. The bladelike structures of the corallite are called _____.
 - a. sclerosepta *
 - b. nematocysts
 - c. zooxanthellae
 - d. gastrovascular cavities
 - e. None of the above
5. Cnidocytes possess small stinging capsules called _____ that are used by the corals to capture zooplankton for food.
 - a. nematocysts *
 - b. sclerosepta
 - c. zooxanthellae
 - d. gastrovascular cavities
 - e. corallites
6. Coral reefs that resemble a ring which rises out of deep water far from land are known as _____.
 - a. atolls *
 - b. barrier reefs
 - c. fringing reefs
 - d. patch reefs
 - e. None of the above
7. _____ reefs occur near land, with a channel of water between the reef and the land.
 - a. Barrier *
 - b. Atoll
 - c. Lagoon
 - d. Patch
 - e. None of the above
8. Coral reefs are about _____ times more productive than the open tropic seas.
 - a. 1,000
 - b. 100 *
 - c. 10
 - d. 5
 - e. 2
9. The amount of plankton available to corals is sufficient to meet about _____% of their total energy needs.
 - a. < 1
 - b. 5–10 *
 - c. 50–60
 - d. 75–80
 - e. 95–99
10. Which one of the following terms provides a correct description for an increase in colony size?
 - a. Budding *
 - b. Fragmentation
 - c. Sexual reproduction
11. Which one of the following terms is the name for the biological process that results in a planula larva?
 - a. Budding
 - b. Fragmentation
 - c. Sexual reproduction *
12. Which one of the following organisms is not a member of the so-called cryptofauna?
 - a. Algae *
 - b. Bryozoans
 - c. Sponges
 - d. Tunicates
13. The response of some corals to aggression by neighbors is to produce very long tentacles called _____ tentacles. These structures possess powerful batteries of nematocysts.
 - a. sweeper *
 - b. pecking
 - c. exploitative
 - d. interference
 - e. None of the above
14. The dominant organisms of coral reefs are _____.
 - a. scleractinian corals *
 - b. ahermatypic corals
 - c. macroalgae
 - d. jellyfish
 - e. predatory gastropods

15. The term ____ describes those tropical and subtropical inshore ecosystems which are dominated by trees that can grow in salt water.
 - a. mangals *
 - b. reefs
 - c. pneumatophores
 - d. sere
 - e. None of the above
16. Sexual reproduction in mangroves occurs by which one of the following mechanisms?
 - a. Viviparous embryos *
 - b. Prop roots
 - c. Pneumatophores
17. In mangroves, the structures sent up above the surface by shallow roots are called _____.
 - a. pneumatophores *
 - b. prop roots
 - c. adventitious roots
 - d. tap roots
 - e. None of the above
18. An organism from which of the following taxonomic groups damages mangrove roots by boring into them?
 - a. Isopoda *
 - b. Gastropod
 - c. Cirripedia (barnacles)
 - d. Crustacea (crabs)
 - e. Bivalvia
19. According to Davis (1940), the first seral stage in mangrove succession is established by which of the following species?
 - a. Red mangrove *
 - b. White mangrove
 - c. Black mangrove
20. Which of the following factors or organisms produces no damage to mangals?
 - a. The hermit crab *Clibanarius* *
 - b. The isopod *Sphaeroma*
 - c. Typhoons and hurricanes
 - d. Herbicide application
 - e. Filling and dredging waters
21. Coral reefs are unique among major marine associations because _____.
 - a. they are formed entirely by biological action *
 - b. they occur everywhere in the world
 - c. they contain no other organisms but corals
 - d. corals are the only organisms to secrete calcium carbonate
 - e. None of the above
22. Zooxanthellae are symbiotic ____ living in corals.
 - a. algae *
 - b. bacteria
 - c. parasites
 - d. protozoans
 - e. viruses
23. The most accepted theory as to the origin of atolls is called _____.
 - a. the subsidence theory *
 - b. the bank theory
 - c. the emergence theory
 - d. the flooding theory
24. Which of the following is a serious disease caused by eating tropical fishes with toxins in their body?
 - a. Ciguatera *
 - b. Crinotoxin
 - c. The bends
 - d. Emphysema
 - e. Squeeze
25. When comparing the production of coral reefs to that of offshore tropical waters, we find that the coral reefs are ____ productive.
 - a. more *
 - b. less
 - c. equally
26. Water and debris leave a coral reef through which one of the following structures?
 - a. Surge channels *
 - b. Lagoons
 - c. Algal ridges
 - d. Reef flats

168 Test Bank for Marine Biology

27. Reefs in the Atlantic rest on shallow platforms which are the result of the erosion that took place in the ____ period.
- Pleistocene *
 - Cambrian
 - Jurassic
 - Permian
28. Small, colonial, encrusting invertebrates, such as sponges, brachiopods, and tunicates, that occupy the undersides of coral heads and cliffs compose a group called the ____.
- cryptofauna *
 - grazing fauna
 - garden fauna
 - polyp
 - planula
29. A specialized form of predation practiced by certain small fish and some shrimps in which ectoparasites are removed from larger fish is known as ____.
- cleaning behavior *
 - ciguatera
 - phase-shift behavior
 - pioneer-species behavior
 - keystone-species behavior
30. *Acanthaster planci* is a ____ noted for its ability to destroy whole coral colonies.
- starfish *
 - cnidarian
 - fish
 - crab
 - clam
31. Mangroves are ____.
- facultative halophytes *
 - obligate halophytes
 - obligate freshwater plants
 - None of the above
32. A catastrophic coral mortality known as coral bleaching results when corals ____.
- expel their zooxanthellae *
 - compete for space by overgrowing one another
 - compete for space by killing each others' tissues
33. The ____ attempts to explain the high species diversity of coral-reef fishes by noting that the production of many offspring by any one type of these fish depends on chance.
- lottery model *
 - competition model
 - predation-disturbance model
 - recruitment-limitation model
34. The algal ridge is coral free because ____.
- the ridge receives the full impact of the wave energy of the surf *
 - the algae living there posses toxic substances that inhibit coral growth
 - there is too much sunlight there
35. Coral reefs are estimated to occupy ____ % of the area of the planet.
- 0.2 *
 - 2
 - 20
 - 50
 - 75
36. Coral reefs are limited to waters of full salinity that are not deeper than 50–70 m, do not fall below ____°C, and do not have excessive sediment.
- 18 *
 - 81
 - 4
 - 100
 - 25
37. Which one of the following is not one of the four competing hypotheses that attempt to explain the high local fish diversity of reefs?
- Uniformity hypothesis *
 - Lottery hypothesis
 - Predator-disturbance model
 - Competition model
 - Recruitment-limitation model
38. Coral reefs have the greatest higher taxonomic diversity of any marine ecosystem, comprising perhaps ____% of all species.
- 4–5 *
 - 10–15
 - 25–30
 - 75
 - 95

39–48. Using the given numbered diagram, identify the following regions: *[Instructors will have to reproduce Figure 9.20 from the textbook.]*

_____ algal ridge	[3 *]
_____ lagoon beach	[6 *]
_____ lagoon reef	[8 *]
_____ leeward reef flat	[9 *]
_____ leeward reef margin	[10 *]
_____ leeward reef slope	[11 *]
_____ outer seaward slope	[1 *]
_____ reef flat	[4 *]
_____ seaward beach of island	[5 *]
_____ windward reef margin	[2 *]

True/False

Mark each of the given statements with the letter *T* if it is true and *F* if it is false.

[Instructors: In some cases, a few options are included for the same question.]

1. [False] Ahermatypic corals are corals that produce reefs.
2. [True] One distinguishing feature between hermatypic and ahermatypic corals is that the former possess zooxanthellae.
3. [False] One distinguishing feature between ahermatypic and hermatypic corals is that the former possess zooxanthellae.
4. [True] Bladelike structures called sclerosepta arise from the base of the corallite.
5. [True] The Indo-Pacific region has many more coral-reef species than the Atlantic region.
6. [False] The Atlantic region has many more coral-reef species than the Indo-Pacific region.
7. [True] Coral reefs of the Pacific are older than those of the Atlantic, which date back only to the Pleistocene period.
8. [True] Corals that use ciliary mucus feeding tend to have shorter tentacles than corals that feed more heavily (directly) from the plankton.
9. [True] Coral bleaching occurs when the symbiotic zooxanthellae of coral polyps are expelled.
10. [True] Mangrove is a term that carries no taxonomic significance because it denotes trees and shrubs composing eight different families.
11. [True] Mangrove is a term without taxonomic significance; rather, it indicates a particular habitat and niche for a variety of flowering plants.
12. [False] Mangrove is a term that has taxonomic significance as well as indicates a particular habitat and niche for a variety of flowering plants.
13. [True] In order to exclude salt and separate freshwater from salt water by the process of reverse osmosis, mangroves must expend energy.
14. [True] The response of some corals to aggression by neighbors is to produce very long tentacles called sweeper tentacles; these structures possess powerful batteries of nematocysts.
15. [True] There is no true seed stage in mangroves such as *Rhizophora*.
16. [False] Recruitment of new coral colonies to a reef is very predictable.
17. [False] Herbivory and predation are more intense on Atlantic reefs than on Pacific reefs.
18. [True] Phototrophic corals are dependent on zooxanthellae for their nutrition.
19. [True] Bacteria are very abundant on coral reefs.
20. [False] The inside of the calcium carbonate cup that a coral polyp sits in is smooth and lacks any ridges.

170 Test Bank for Marine Biology

21. [False] The Great Barrier Reef is actually misnamed, for it is really an atoll.
22. [True] Coral reefs are among the most productive systems in the marine environment and have existed for hundreds of millions of years; yet they are sensitive to a number of natural and human-made disturbances.
23. [False] Corals are members of the phylum Porifera.
24. [True] Although corals are the major organisms that form reef structures, members of practically all phyla and classes may be found on coral reefs.
25. [False] Reef corals are carnivores; thus, they depend upon their prey for all of their nutrition.

Short Answer (Fill in the Blank)

1. New colonies of coral are established by fragmentation of skeletal pieces or by settlement of a subsequent development of a planktonic larva called a planula.
2. If a surge channel is overgrown by a roof of algae, the surge channel can end in a blowhole that resembles a geyser.
3. Coral reefs of the Atlantic Ocean date only from the Pleistocene period.
4. Besides augmenting the energy supply for hermatypic corals, the zooxanthellae help in calcification.
5. Lunar events regulate the release of eggs and sperm in coral species called broadcast spawners.
6. The cryptofauna refers to invertebrates that live in crevices or burrow into the calcium carbonate matrix of coral reefs.
7. The loss of the symbiotic algae from hermatypic corals is called coral bleaching.
8. The uninterrupted development of embryos while still attached to the parent plant in mangroves is called vivipary.
9. Some mangroves possess tissues that act as salt glands, which permits them to secrete excess salt.
10. Mangroves use pneumatophores to bring oxygen into their tissues.
11. Coral polyps occupy small cups called corallites in a massive calcium carbonate skeleton.
12. The bladelike structures of the corallite are called sclerosepta.
13. Fast-growing corals can grow up and over other corals, shutting them out from light and thus killing the shaded corals. This kind of interactions is called exploitative competition.
14. Certain slow-growing corals can kill the tissues of adjacent corals with which they are in competition. This kind of interaction is called interference competition.
15. Corals appear to be able to limit their zooxanthellae density and growth rates through some sort of metabolic growth factor.
16. The symbiotic zooxanthellae of corals are actually members of the planktonic algae group called dinoflagellates.
17. Nematocysts are tiny structures implanted within the tentacles of corals. They are responsible for capturing prey by embedding tiny harpoonlike threads into the prey.
18. The exoskeleton of a coral is composed of calcium carbonate.
19. The greatest human-caused destruction of mangroves took place during the Vietnam War.
20. Coral reefs, salt marshes, and mangrove forests are biogenic structures built by biological activity.
21. Coral reefs may be grouped into three categories: fringing, barrier, and atoll.
22. Besides his theory of natural selection, Charles Darwin also proposed a subsidence theory to explain the formation of coral atolls.

Essay Discussion Questions

Develop complete answers to these questions.

1. Explain Charles Darwin's subsidence theory for the development of coral atolls, and discuss how this idea was tested.
2. Discuss how coralline algae are important to reef formation.
3. How did the Pleistocene ice ages affect coral reefs in the Indo-Pacific and Atlantic Oceans?
4. Examine the ideas that have been suggested which may explain how coral reefs can be so productive.
5. How did the catastrophic mortality of the Caribbean sea urchin, *Diadema antillarum*, affect the coral reefs of that region? Discuss.
6. Besides being important primary producers, discuss other roles algae may play in the coral-reef ecosystem.
7. Review the two types of coral predators (grazers and coral destroyers), and discuss the significance of their impact on the coral reef.
8. Discuss how competitive hierarchies help to maintain species diversity of coral with increasing depth on a reef.
9. How is coral diversity maintained by a series of behaviors termed "aggressive pecking order"?
10. Examine species richness in coral and coral fishes as a function of geography.
11. Speculate as to why toxins are commonly found in coral-reef invertebrates and fishes.
12. Many coral-reef fish are conspicuously colored. Discuss the functions that such an advertisement might serve.
13. Summarize why cryptofauna are important to the coral reef.
14. What sorts of animals make up the so-called cryptofauna, and how are they important to the coral-reef ecosystem?
15. What is *Acanthaster planci*, and how has it influenced the coral-reef ecosystems of the western Pacific Ocean?
16. Review the theories that attempt to explain the population explosion of *Acanthaster planci*. Group them into the categories of natural and artificial occurrences.
17. What evidence suggests that *Acanthaster planci* population explosions have occurred naturally in the past?
18. What is ENSO, and of what significance is it to coral reefs in the Pacific and Caribbean?
19. How does an extensive root system help to stabilize the sediment of a mangal? Discuss.
20. Besides wave action, what other factors are important in determining mangrove distribution? Present evidence that supports your argument.
21. How do mangals significantly change their environment? Support your essay with evidence.
22. Where are coral reefs found in the world's oceans? Explain why they are found there.
23. Explain why the Atlantic coral reefs are only as old as the last North American ice age.
24. Of what do coral reefs consist? What is the major reef-building organism?
25. What are the major reef-destroying factors?
26. Distinguish between hermatypic and ahermatypic corals, and indicate their relative importance to reef growth.
27. What are zooxanthellae? Summarize their importance to coral reefs.
28. What important factors lead to coral death? Consider physical, chemical, and biological factors, and assess their relative importance.
29. Researchers have determined that zooplankton cannot be the sole C/E source of corals. In table form, list all the C/E sources of corals, and briefly review how the coral go about gathering those foods.

172 Test Bank for Marine Biology

30. Discuss Darwin's subsidence theory for the ontogeny of a coral atoll. Under what circumstances did Darwin see coral atolls arising? How was Darwin's theory eventually formulated into a hypothesis and then tested?
31. Explain how guyots provide supporting evidence that Darwin was right about the ontogeny (development) of coral atolls.
32. Explain what Darwin meant by the "coral-reef problem." How was this question finally laid to rest? Who did the hypothesis testing that solved the problem?
33. What are coralline algae, and what role do they play in reef formation? In reef stabilization?
34. Discuss zonation (both horizontal and vertical) in coral reefs.
35. Discuss why the conditions for coral growth within a coral-reef lagoon are not uniform.
36. How has the zooxanthellae–coral polyp symbiosis circumvented the lack of nutrients in the water surrounding tropical reefs? What other features of coral reefs help to boost their productivity?
37. How are zooxanthellae important to CaCO_3 deposition by coral polyps? Discuss.
38. From what you know of the coral-reef ecosystem, photosynthesis, and the chemistry of the carbonate equilibrium, suggest a model that explains how zooxanthellae are important to CaCO_3 deposition.
39. Discuss how microatolls develop.
40. How do coral-reef organisms compete for space? How does aggression play a role in this competition?
41. Why are large mollusks and echinoderms more immune to predation than smaller, soft-bodied invertebrates?
42. How can *Acanthaster planci* provide an opportunity to slow-growing corals to survive?
43. Develop a table that lists five of the organisms responsible for coral-reef destruction, and explain how each organism accomplishes this damage.
44. Why are there so many different types of coral-reef fishes? Why are there fewer species of coral-reef fishes in the Atlantic Ocean than in the Pacific?
45. What sort of diel variation is seen in coral-reef fishes? Of what significance is this variation?
46. Compare and contrast mangals with forests in tropical and temperate climates. Consider species diversity, productivity, and anatomical adaptations.
47. How do the mangals influence development of new land areas?
48. What issues are involved in the question of whether ecological succession occurs on coral reefs?
49. Discuss the kinds of human activities that can and have damaged coral reefs.
50. How might the microbial loop help explain the high productivity of coral reefs?
51. How does the wind influence zonation of a coral reef? Discuss.
52. How is ciguatera a disease that is related to human culture? Discuss.
53. What are the natural causes of coral die-off? Of these causes, which may be influenced by human activities? Discuss.
54. How do mangals change the typical marine mud and sand, making them into a habitat more suitable to many more invertebrates? What sorts of invertebrates are associated with mangals?
55. Discuss the following metaphor: Coral reefs are like canaries brought into a mine.
56. The cryptofauna refers to those invertebrates which inhabit the coral substrates living in cracks, crevices, and caves and under and in coral heads and rock. How are they important as agents of bioerosion?
57. How is vivipary important to the life cycle of mangroves? Discuss.
58. Why is it that although coral reefs occupy less than 0.2% of the area of our planet, they occupy so much human interest?
59. Challenge Question: Support or refute the statement that space is a limiting factor in the marine environment.

60. Challenge Question: Discuss the life cycle of corals, and contrast it with that of jellyfish. What are the consequences for the genetic diversity of local coral populations as compared with the genetic diversity of local jellyfish populations?
61. Challenge Question: Discuss the four models that have been offered to explain the diversity of species among coral-reef fishes.
62. Challenge Question: Speculate on what a microbial-loop system, as is found in the plankton, might look like for a coral reef.
63. Challenge Question: Of what interest are coral reefs to the pharmaceutical industry?
64. Challenge Question: How does the theory of plate tectonics help to explain coral atolls?
65. Challenge Question: Compare the various zones of a coral-reef with a rocky coastline containing tide pools.
66. Challenge Question: Why are mangroves shallowly rooted, and not deeply rooted, plants?
67. Challenge Question: How have corals been used in human reconstructive surgery?
68. Challenge Question: Discuss the ecosystem services of mangrove forests. *[Instructors will have to make sure that the topic of ecosystem services is covered in class.]*
69. Challenge Question: Discuss the ecosystem services of coral reefs. *[Instructors will have to make sure that the topic of ecosystem services is covered in class.]*
70. Challenge Question: Summarize how paleoclimatic data of sea surface temperature can be recovered from the ratio of Sr to Ca present in the coral skeletal matrix. *[Instructors: Refer students to the following reference as an outside reading:*
Beck, J.W., et al. 1992. Sea-surface temperature from coral skeletal Strontium/Calcium ratios. *Science* 257: 644–647.]
71. Challenge Question: Compare and contrast the dynamics of coral-reef formation with the formation of a glacier.
72. Challenge Question: In light of Robert Frost's poem *Tuft of Flowers*, discuss the test of Darwin's subsidence theory. (N.B.: With apologies to the women in the class; remember that Frost wrote this poem at a much different time.) *[Instructors will have to supply the poem.]*

